DAT32-2

Exploratory Data Analysis & Data Quality

Albert School — 16 hours

Contents

Preface

In **Pandas I** (DAT12-3) you learned to take a raw CSV and turn it into a clean, summarised DataFrame: to inspect it with `head`, `info` and `describe`, to filter with `loc` and boolean masks, to group and aggregate, to merge tables, and to write a paragraph interpreting what you found. **Scraping, Pandas II & Algorithmic Thinking** (DAT22-1) added window functions, time-indexed frames, and the habit of collecting your own data and forming hypotheses about it. **SQL & Data Querying** (DAT22-5) taught you to translate a business question into a query, to join and aggregate relational data, and to spot quality problems — duplicates, NULLs, inconsistent strings — at the query level.

You can therefore already **manipulate** data. This course is about what you do **before** you reach for a model and **around** the manipulation you already know: how to turn a vague business situation into a question worth answering, how to interrogate a dataset systematically rather than poking at it, how to judge whether the data is trustworthy enough to support a conclusion, and how to report what you found so that someone who cannot read code can act on it. The single discipline this course installs is restraint: **look honestly before you conclude, and conclude only what the data can bear.**

A note on method. Every idea arrives as a concrete example first and the general statement second. We keep one running case throughout so that each new tool attaches to numbers you already half-know.

> **The running case — FreshCart.** FreshCart is a meal-kit subscription company: customers sign up, receive weekly boxes, and may cancel at any time. Management has noticed that **monthly churn is rising** and wants to know why. Our dataset has one row per customer with columns such as `signup_date`, `region`, `plan` (Basic / Premium), `weekly_orders`, `avg_box_value`, `delivery_delay_days`, `satisfaction` (a 1–5 survey score, often missing), and `churned` (yes/no). We will scope this situation into a question, profile the data, explore it, and write the report — in that order.

1 From a business situation to a data question

Management's brief is: *"Churn is rising — find out why and fix it."* That is a **business situation**, not a question you can answer with data. "Find out why" names no quantity, no population, and no standard for being done. If you open a notebook now, you will explore aimlessly and produce a deck nobody can act on. The first deliverable of any project is therefore not a chart but a **scoped question**.

> **Definition 1** **Scoping** is the translation of a vague business situation into a **data question**: a question that names
> - the **quantity** to be measured or the **relationship** to be investigated,
> - the **population** it concerns (which rows, over which period),
> - the **data** required to answer it, and whether that data exists and is accessible.

> **Worked example — scoping the FreshCart brief.** "Find out why churn is rising" becomes, after scoping:
>
> *"Among customers who signed up in the last 12 months, which of {plan, region, delivery delay, early-weeks satisfaction} is most strongly associated with cancelling within 90 days of signup, and by how much?"*
>
> This names a quantity (90-day churn rate), a population (last-12-months signups), candidate drivers, and the columns required. Two analysts handed this question would agree on whether a given result answers it. "Find out why" fails every one of those tests.

Scoping forces three decisions to be made **explicitly and in writing**, before any analysis:

- **What data is needed** — and does it exist, at the right grain, accessibly? If satisfaction is only collected from a self-selected few, that limits what the question can claim.
- **What "good enough" looks like** — the accuracy, coverage, or confidence at which the stakeholder would actually act. A direction ("Premium churns less") may be enough to decide; a precise elasticity may not be needed.
- **What is out of scope** — the questions you will *not* answer. "We will describe associations, not prove causes" is a scope boundary that saves you from promising more than EDA can deliver.

> **Common pitfall** The most common scoping failure is the **implicit causal promise**. "Find out why churn is rising" invites you to deliver causes, but exploratory analysis yields **associations**. Scope the question as associational ("which factors are associated with churn") and state explicitly that establishing cause would require an experiment. Promising "why" and delivering "what correlates with" is how analyses lose a stakeholder's trust.

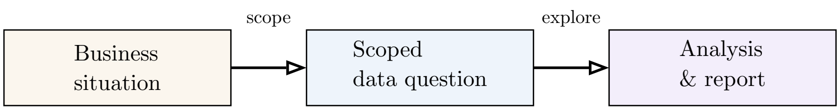


Figure 1: The project always flows left to right. Most failed analyses skip the middle box and jump straight from a vague situation to charts.

2 Systematic exploratory data analysis

With the question scoped, you explore. The temptation is to plot whatever comes to mind; the discipline is to proceed in a fixed order, from one variable to many, so that nothing is overlooked. **Exploratory data analysis** (EDA) is that systematic first look.

Worked example — the natural order of looking. To understand FreshCart churn you would, in order: (1) look at each column on its own — how is `weekly_orders` distributed? what fraction `churned`? — then (2) pair churn against each candidate — do Premium customers churn less? — then (3) look at several at once — does the delay–churn link hold *within* each region? At every step you also watch for records that simply do not belong. That ordered sweep is exactly what EDA formalises.

Definition 2

- **Univariate analysis** examines one variable at a time: its distribution, central tendency, spread, and shape.
- **Bivariate analysis** examines the relationship between two variables.
- **Multivariate analysis** examines three or more variables simultaneously.
- **Anomaly detection** is the identification of observations — values, records, or patterns — that depart markedly from what the rest of the data follows.

The order matters. Univariate analysis tells you what is normal for each column, which is what lets you recognise an anomaly at all; bivariate analysis surfaces candidate relationships; multivariate analysis is where those relationships are confirmed, qualified, or **overturned** (see Simpson's paradox, later). Skipping to bivariate plots before profiling each variable is the most common way to be fooled by a value that was an error all along.

The EDA loop, per variable and per pair: describe → visualise → flag anomalies → form a hypothesis → carry it to the next level of analysis. EDA is iterative, not a single pass.

3 Choosing the right visualization

A chart is an instrument for answering one kind of question; using the wrong one is like reading a thermometer to find your weight. Before choosing a chart, name the question type.

Worked example — four questions about FreshCart, four charts.

- *How are box values spread?* → a **histogram** of `avg_box_value`.
- *Do delivery delays relate to churn?* → a **scatter** (or box plot of delay, split by churned).
- *What share of revenue comes from each plan?* → a **bar chart** of revenue by plan.
- *How has weekly churn moved this year?* → a **line chart** over time.

Each question type has a natural chart; mismatches mislead.

Definition 3 The four EDA question types and their standard charts:

- **Distribution** (the shape of one variable): histogram, box plot, density plot.
- **Relationship** (how two variables move together): scatter plot, with a line of best fit.
- **Composition** (how a whole divides into parts): bar chart, stacked bar, occasionally a pie chart.
- **Time trend** (how a variable moves over time): line chart.

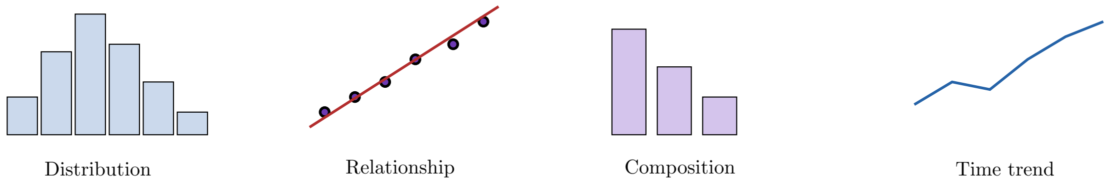


Figure 2: Match the chart to the question. A distribution question is not answered by a composition chart, and vice versa.

Common pitfall Two recurring chart mistakes: using a **pie chart** with many slices (the eye cannot compare angles — a bar chart almost always reads better), and using a **line chart for a non-temporal x-axis** (a line implies continuity and ordering between points; reserve it for time or another genuinely ordered axis).

4 Missing data mechanisms

In FreshCart, `satisfaction` is blank for many customers. Your instinct from Pandas I is to impute it with the mean. **Stop.** Whether that is legitimate depends entirely on *why* the value is missing — and there are three different reasons, with three different consequences.

Worked example — three reasons satisfaction is blank.

- The survey tool randomly sampled 30% of customers to ask. Missingness has nothing to do with anything — **MCAR**.
- The survey was only emailed to customers on the Premium plan; within each plan, who answered is unrelated to their happiness. Missingness depends on `plan`, an **observed** column — **MAR**.
- Everyone was asked, but **unhappy customers ignored the survey**. Whether the score is missing depends on the score itself — **MNAR**. This is the dangerous one, and for churn analysis the most likely.

Definition 4

- **MCAR** (Missing Completely At Random): the probability of being missing is unrelated to any data, observed or not.
- **MAR** (Missing At Random): the probability of being missing depends only on **observed** variables, not on the missing value itself.
- **MNAR** (Missing Not At Random): the probability of being missing depends on the **unobserved** missing value itself.

The mechanism dictates the strategy:

- Under **MCAR**, dropping the affected rows loses data but does not bias what remains; simple imputation is also safe.
- Under **MAR**, imputation **conditioned on the observed variables** (e.g. impute satisfaction within each plan) is justified; a global mean is not.
- Under **MNAR**, both dropping and naive imputation **bias the result** — here, imputing the mean would make dissatisfied churners look averagely happy, hiding the very signal you are chasing. MNAR must be modelled explicitly or disclosed as a limitation.

Common pitfall You cannot fully prove the mechanism from the data alone — MNAR in particular is invisible to the dataset because the informative values are exactly the ones you do not have. Mechanism is argued from **how the data was collected**, not read off a table. When in doubt between MAR and MNAR, treat it as MNAR and disclose it; the cautious error is cheaper.

5 Outlier handling

`describe()` shows a customer with `weekly_orders` of 480 and another with `delivery_delay_days` of −3. Both are outliers, but they call for opposite actions — and a third kind of outlier calls for no action at all. An extreme value is **not** automatically an error.

Worked example — three outliers in FreshCart.

- `delivery_delay_days = -3`: a delivery cannot arrive before it ships. This is a **data-entry error** — fix it or remove it.
- `weekly_orders = 480`: real, but it is an office that pools its orders, not a household. A **legitimate extreme** — keep it, but perhaps analyse business accounts separately.
- `avg_box_value = 210` for a few gourmet-tier customers: simply the long right tail of a skewed price distribution. **Distribution behaviour** — keep it, and acknowledge the skew (and consider a log transform — see feature engineering).

Definition 5 An outlier may be:

- a **data-entry error** (an impossible or corrupted value) — correct or remove;
- a **legitimate extreme** (a real but unusual observation) — keep, possibly segment;
- **distribution behaviour** (an expected tail value of a skewed or heavy-tailed distribution) — keep, and report the shape.

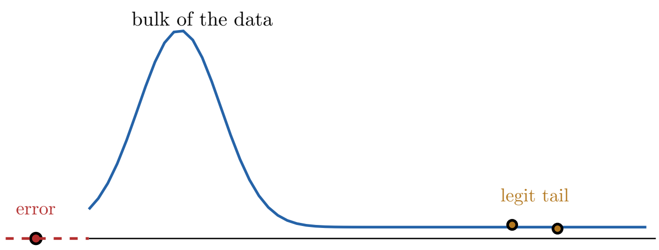


Figure 3: Not every extreme is an error. The point left of zero is impossible (a data-entry error); the far-right points are genuine tail values of a skewed distribution and must be kept.

Common pitfall Do not delete outliers just because a rule (e.g. “beyond 3 standard deviations”) flags them. Such rules detect **candidates**; the classification — error vs. legitimate vs. tail — is a judgement about the world, not a threshold. Deleting legitimate extremes manufactures a tidy dataset and a wrong conclusion.

6 Correlation analysis

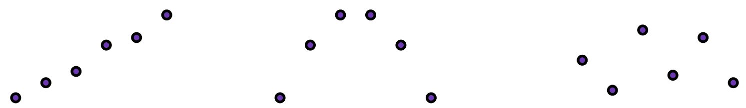
Bivariate exploration of FreshCart’s numeric columns is summarised compactly by a **correlation matrix**: a table of pairwise linear associations, each between −1 and +1.

Definition 6 The **(Pearson) correlation** between two numeric variables measures the strength and direction of their **linear** association, on a scale from -1 (perfect negative) through 0 (none) to $+1$ (perfect positive). A **correlation matrix** is the square table of pairwise correlations among a set of variables; its diagonal is always 1 and it is symmetric.

You read a correlation matrix by scanning off-diagonal entries for values near ± 1. Suppose FreshCart's matrix shows `delivery_delay_days` and `churned` correlated at $+0.61$ — strong and positive. It is tempting to conclude "delays cause churn, so fix logistics."

Common pitfall **Correlation does not imply causation.** Two variables can move together because one causes the other, because a **third variable** drives both, or by sheer coincidence. The textbook example: ice-cream sales and drowning deaths are strongly positively correlated, yet neither causes the other — hot weather drives both. In FreshCart, delays and churn may both be driven by a struggling regional depot; "fix delays" might not move churn if the depot's real problem is product quality.

Worked example — two cautions reading the matrix.
- Correlation captures only **linear** association. A variable that rises then falls (e.g. churn vs. tenure) can have near-zero correlation while being strongly related — always pair the matrix with a scatter plot.
- A correlation near 0 means "no *linear* trend," not "no relationship."



$r \approx +0.9$ $r \approx 0$ (but related!) $r \approx 0$ (none)

Figure 4: Correlation measures **linear** association only. The middle cloud has a strong relationship that correlation reports as zero — which is why a matrix is never read without scatter plots.

7 Feature engineering basics

Raw columns are not always the most informative form of the data. **Feature engineering** derives new columns that make structure easier to see and to model.

Worked example — four engineered features for FreshCart.
- **Binning**: turn `age` into "18–29 / 30–44 / 45–59 / 60+" to compare churn across life stages.
- **Log transformation**: replace the right-skewed `avg_box_value` with log(avg_box_value) so the gourmet tail stops dominating plots and the relationship with churn becomes roughly linear.
- **Interaction term**: `is_premium times high_delay` — the effect of a delay may be far worse for Premium customers, an effect neither column shows alone.
- **Date feature extraction**: from `signup_date`, derive `signup_month`, `signup_dayofweek`, `is_weekend`, and `tenure_days` — far more useful than a raw timestamp.

Definition 7

- **Binning**: converting a continuous variable into discrete ordered intervals.
- **Log transformation**: replacing a variable by its logarithm to compress a skewed range and linearise multiplicative relationships (defined for positive values only).
- **Interaction term**: a new feature built from the product or combination of two features, capturing an effect that depends on both jointly.
- **Date feature extraction**: deriving calendar features (day of week, month, quarter, is-weekend, elapsed tenure) from a date or timestamp.

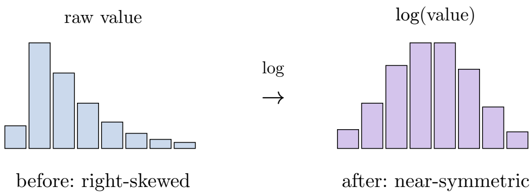


Figure 5: A log transformation pulls in a long right tail, turning a skewed distribution into a roughly symmetric one — easier to plot, summarise, and relate linearly to other variables.

Common pitfall log(0) and log(negative) are undefined, so a column with zeros or negatives cannot be log-transformed directly — analysts use $\log(x+1)$ (the `log1p` function) for non-negative data with zeros. And binning always discards information: choose bin edges from the domain (life stages, price tiers), not arbitrarily, or you can manufacture or hide a pattern.

8 Data quality reporting

Before any finding can be trusted, the dataset itself must be profiled. A **data quality report** assesses the data along four standard dimensions and states, for each, what is wrong and how it affects the analysis. This formalises, at the dataset level, the quality checks you met at the query level in SQL.

Worked example — FreshCart's quality report in miniature.

- **Completeness**: `satisfaction` is 58% missing (and likely MNAR — see above); every other column is over 99% present.
- **Consistency**: `region` appears as both "IDF" and "Île-de-France"; three rows have `churned = no` but a non-empty `churn_date`.
- **Uniqueness**: 412 rows share a `customer_id` with another row — duplicate signups that would double-count churn.
- **Timeliness**: the export is six weeks old, so "last-90-days churn" is really "churn as of six weeks ago" — material for a metric that moves weekly.

Definition 8 The four data-quality dimensions:

- **Completeness**: the extent to which required values are present (the inverse of missingness).
- **Consistency**: the absence of contradictions within and across records — the same entity recorded the same way, values that agree with one another.
- **Uniqueness**: the absence of unintended duplicate records.

- **Timeliness**: whether the data is current enough for the question it must answer.

A good report does not merely list defects; for each it states the **impact** on the scoped question and the **remedy** applied (deduplicated, standardised, disclosed). A quality issue that does not touch the question can be noted and set aside; one that does — like the MNAR satisfaction gap — must shape what the analysis is allowed to claim.

9 Hypothesis formulation

EDA's job is not to settle questions but to **generate sharp ones**. Each pattern you notice becomes a documented, testable hypothesis with a proposed way to test it.

> **Worked example — from observation to documented hypothesis.** You observe in the data that Premium customers in delayed regions churn far more than Premium customers elsewhere. Documented:
>
> *H: Among Premium customers, churn rises with delivery delay more steeply than among Basic customers (a delay $\times$ plan interaction).* **Approach:** compare 90-day churn across delay quartiles, split by plan; if the question becomes confirmatory, test the interaction term in a model.

Definition 9 A **data-driven hypothesis** is a specific, testable statement about the data, suggested by the EDA, stated precisely enough to be checked. Each is documented with a proposed **analytical approach**: the data, comparison, or method that would confirm or refute it.

A complete EDA deliverable documents **at least five** such hypotheses, each paired with its approach. Writing the approach alongside the hypothesis is what keeps the list honest: a hypothesis you cannot say how to test is not yet a hypothesis.

10 Simpson's paradox

This is the deepest warning in the course, and the clearest reason multivariate analysis is not optional. A relationship that holds in the aggregate can **reverse** inside every subgroup.

> **Worked example — the discount that 'reduces' churn.** FreshCart ran a discount campaign. Aggregated, discounted customers churned **less** than non-discounted ones — so the discount looks like it works. But the discount was offered mostly in the company's **healthy** regions, which churn little anyway. Split by region, in **every single region** the discounted customers actually churned **more** (price-sensitive bargain-hunters). The aggregate verdict is the exact opposite of the truth, and acting on it would scale a campaign that increases churn.

Definition 10 **Simpson's paradox** occurs when a trend present in aggregated data reverses or disappears when the data is split into subgroups. It arises when a **confounding variable** is distributed unevenly across those subgroups.

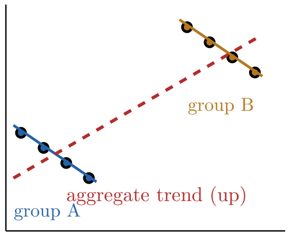


Figure 6: Each subgroup trends **downward**, yet the pooled data trends **upward**. The reversal is driven by the uneven placement of the groups — the confounder. Only disaggregation reveals the truth.

The defence is procedural: before reporting any aggregate relationship, **split it by the obvious confounders** (here, region) and check the relationship still holds inside each subgroup. If it reverses, the subgroup story is the real one.

11 Analytical communication

The investigation is worthless until someone acts on it, and the person who acts usually cannot read your notebook. The final deliverable is a written report a non-technical stakeholder can use.

> **Worked example — the FreshCart report in one breath.** *"90-day churn has risen from 8% to 12%, concentrated in three regions with a failing depot (chart 1). Delivery delay is the strongest correlate of churn there; the recent discount campaign is associated with **higher**, not lower, churn once region is accounted for (chart 2 — Simpson's paradox). Satisfaction data is 58% missing and skewed toward the unhappy, so treat satisfaction findings as suggestive only. Recommendation: fix the three depots before expanding discounts; instrument satisfaction collection so it is no longer self-selected."* — question, finding, limitation, recommendation, in order.

Definition 11 An analytical report communicates through:

- a **narrative arc**: the question, what was found, and what it means, in an order a non-technical reader can follow;
- **visualization selection**: charts chosen to support each finding, matched to the question as in the EDA;
- **limitations disclosure**: an honest statement of what the data and analysis cannot support (missingness, lack of causal proof, staleness);
- **recommendations**: the specific actions the findings support.

Common pitfall The hardest discipline in reporting is **limitations disclosure**. Burying the MNAR satisfaction gap or the correlation-is-not-causation caveat makes the report look stronger and be weaker. State limitations plainly and **next to the finding they qualify**, not in a footnote — it is what separates an analyst a stakeholder can trust from one they cannot. The report is judged by a single test: can the intended audience act on it correctly without your code?

The arc of the whole course, in one line per stage: scope the question → profile quality → explore univariate, then bivariate, then multivariate → handle missingness and outliers by mechanism, not reflex → read correlations without inferring cause → engineer clarifying features → disaggregate against Simpson's paradox → document hypotheses → report with limitations disclosed.

Exercises

1. Take a vague business situation (e.g. "our app's ratings are dropping") and scope it into a specific, answerable data question. State the quantity, the population, the data needed, what "good enough" means, and what is out of scope.
2. Design and execute a systematic EDA on a dataset of your choice: univariate, then bivariate, then multivariate analysis, with anomaly detection at each stage. Explain why the order matters.
3. For four findings, choose an appropriate visualization (distribution, relationship, composition, time trend) and justify each choice. Then give one example of a chart–question mismatch and say what it would mislead a reader to believe.
4. For a dataset with missing values, argue from the collection process whether each missing column is MCAR, MAR, or MNAR, and justify an imputation or exclusion strategy for each. Identify which column is most likely MNAR and why naive mean imputation would bias the result.
5. Detect outliers in a numeric variable and classify each as a data-entry error, a legitimate extreme, or distribution behaviour. Decide how to handle each and explain why a blanket "drop beyond 3σ" rule would be wrong here.
6. Compute and interpret a correlation matrix. Identify the strongest pair, then give a concrete example of two variables that are correlated but not causally related, naming a plausible confounder.
7. Apply binning, log transformation, an interaction term, and date-feature extraction to a dataset. For the log transform, handle any zeros correctly and show the before/after distribution.
8. Profile a dataset end-to-end and write a data-quality report covering completeness, consistency, uniqueness, and timeliness. For each defect, state its impact on a scoped question and the remedy applied.
9. From your EDA, formulate and document at least five data-driven hypotheses, each paired with a concrete analytical approach to test it.
10. Construct or find a real case of Simpson's paradox. Show the aggregate relationship, the reversed subgroup relationships, and name the confounder. Explain why acting on the aggregate would be a mistake.
11. Write a one-page analytical report for a non-technical audience that communicates findings, supporting visualizations, limitations, and recommendations — placing each limitation next to the finding it qualifies.